

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any one of the following.

- A) Prove that total complex power fed into a volume is equal to the algebraic sum of the active power dissipated as heat, plus the reactive power proportional to the difference between time-average magnetic and electric energies stored in the volume, plus the complex power transmitted across the surface enclosed by the volume. 12
- B) Prove that Skin Depth $\delta = \frac{1}{\alpha} = \frac{1}{\beta}$ 12

Q.2 Discuss in details Any Two points of the following.

- A) Radiation Intensity 6
- B) Effective Area 6
- C) Field of an Antenna 6

Q. 3 Design and explore Any one of the following.

- A) Design and array of eight elements spaced at $\lambda/2$ distance 12
- B) Explain the concept of pattern multiplication in detail 12

Q.4 Illustrate Any one array of the following.

- A) Explain the Dolph - Chebychev method of array synthesis 12
- B) Obtain the Tschebyshev Polynomial corresponding to $m = 7, 8, 9$ 12

Q. 5 Design Any two antenna of the following.

- A) Rhombic Antenna 6
- B) Yagi Uda Antenna 6
- C) Log Periodic Antenna 6

*** End ***